# NMC summary

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## Table of contents

**Summary** **1**
Data cleaning . . . . . 1
Exploring the data . . . . . 2
Fix rates . . . . . 2
Steps . . . . . 3

## Summary

### Data cleaning

At this point, I've cleaned both YT and BC data to:

1) remove points that are not geographically feasible (in Germany, the ocean, etc)

2) remove any pre-/post-deployment points

3) remove first week of data (capture effect)

4) remove low quality steps: DOP $> 11$ (if recorded)

5) remove duplicate points

6) remove unreasonably fast steps (preliminary)

7) remove fast round trips

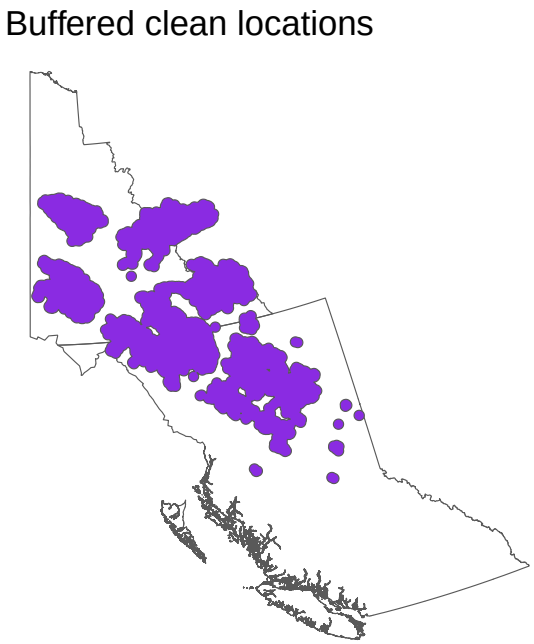
8) remove defunct clusters (post death/fall off)

9) remove individuals with less than 3 points

Now, we have a "clean" data set.

## Exploring the data

Now we're going to explore the clean data.




Clearly, still not perfectly clean, but it's a good starting point.

At this point, there are 1223218 in our cleaned data set.

### Fix rates

Now we're going to explore what the range of fix rates looks like across the whole data set

Quantiles of mean fix rates by individual

```
       0%        25%        50%        75%       100%
 1.062915   5.841326   9.129221  14.907460 573.909190
```

Quantiles of median fix rates by individual

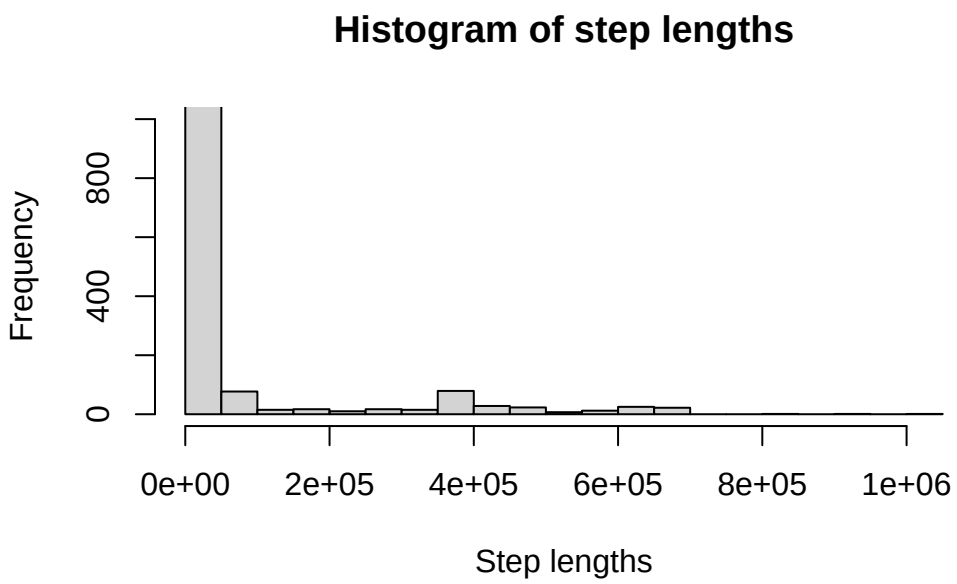
```
   0%    25%    50%    75%   100%
 1.00   5.75   8.00  13.00 240.50
```

Mean of the median fix rates by individual

```
[1] 11.02585
```

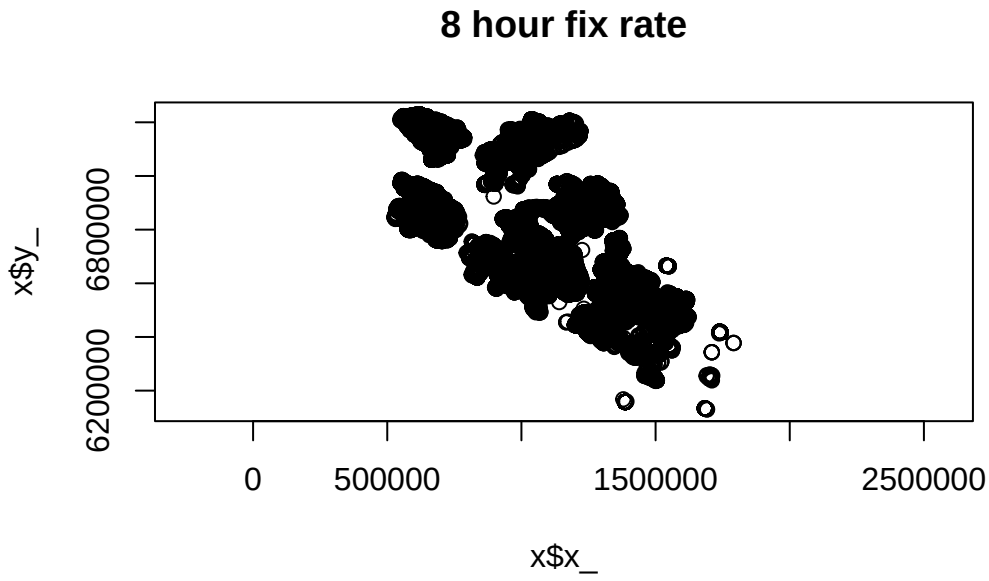
### Steps

Let's explore the all the step regardless of fix rate

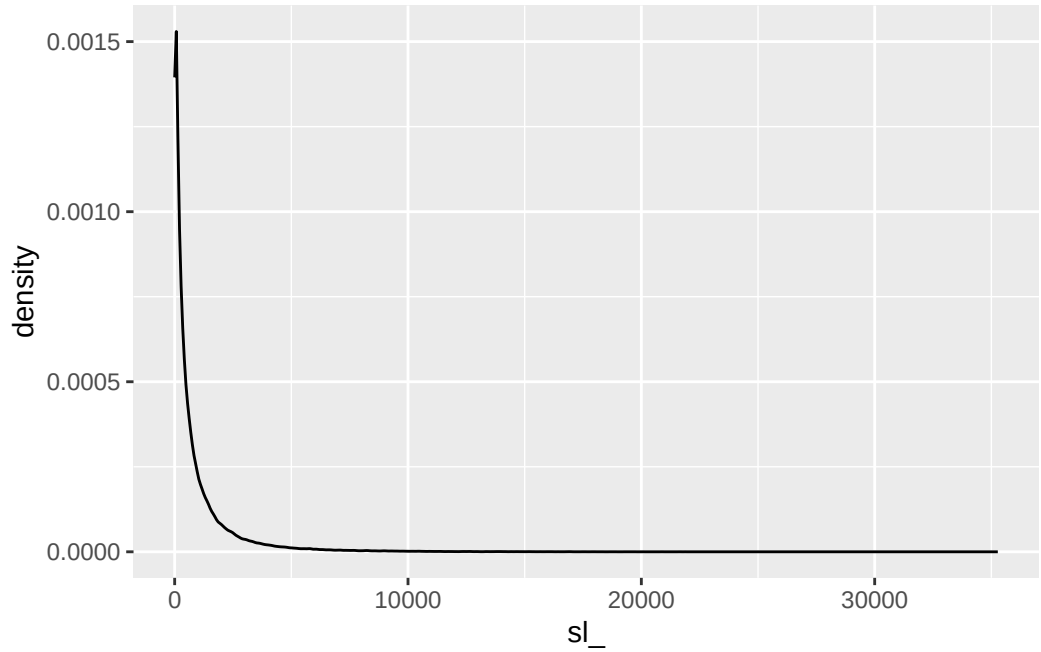



### 8 hour fix rate

The median of median fix rates was 8 hours, so let's check out resampling to this fix rate




Step length distribution




This is a typical distribution, though it looks like there are still a few unnaturally long steps. So let's check the quantiles.

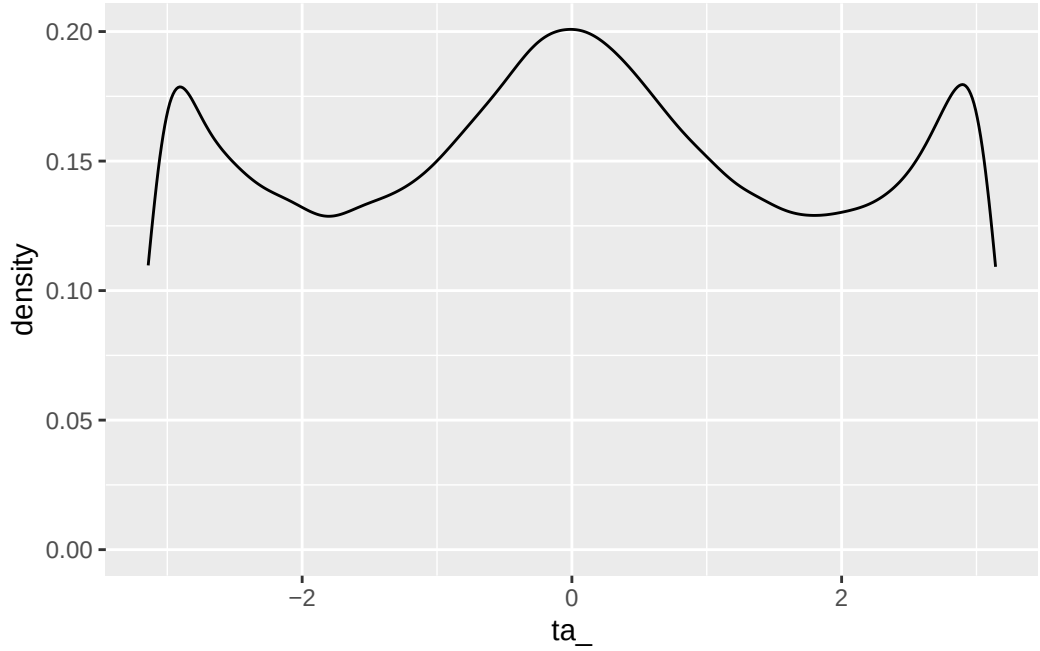
```
       0%          25%          50%          75%         100%
  0.00000     88.30026    366.36013   1069.28278 35276.71324
```

At the extremes:

```
     90%       95%       99%
2412.400 3784.069 8145.762
```

So, looks like we could filter out the top 99% of steps if we need to remove the unrealistically long 34 km step taken in within 8 hours…

Now, lets check out the turn angles

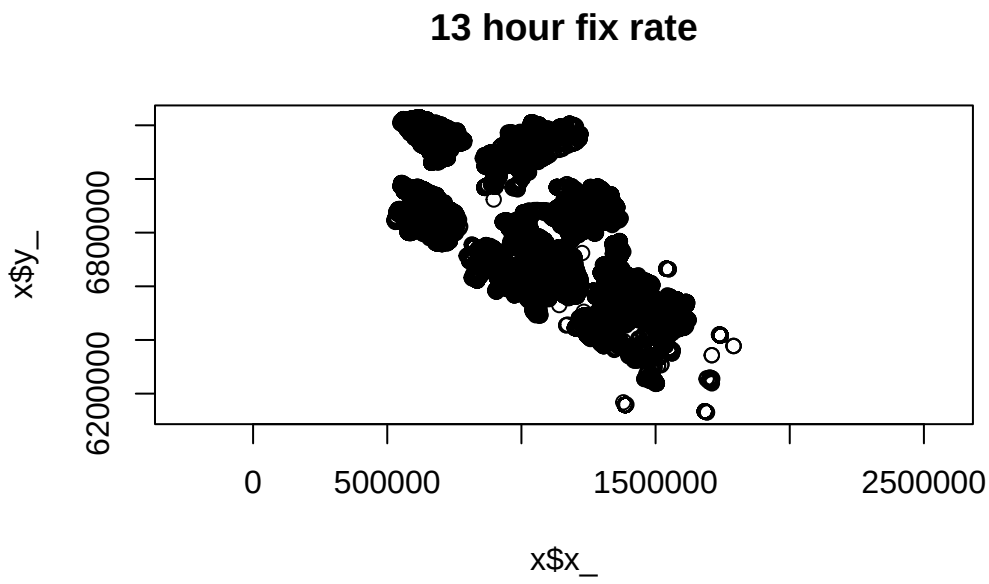



That's a nice looking von Mises distribution, so looks good.

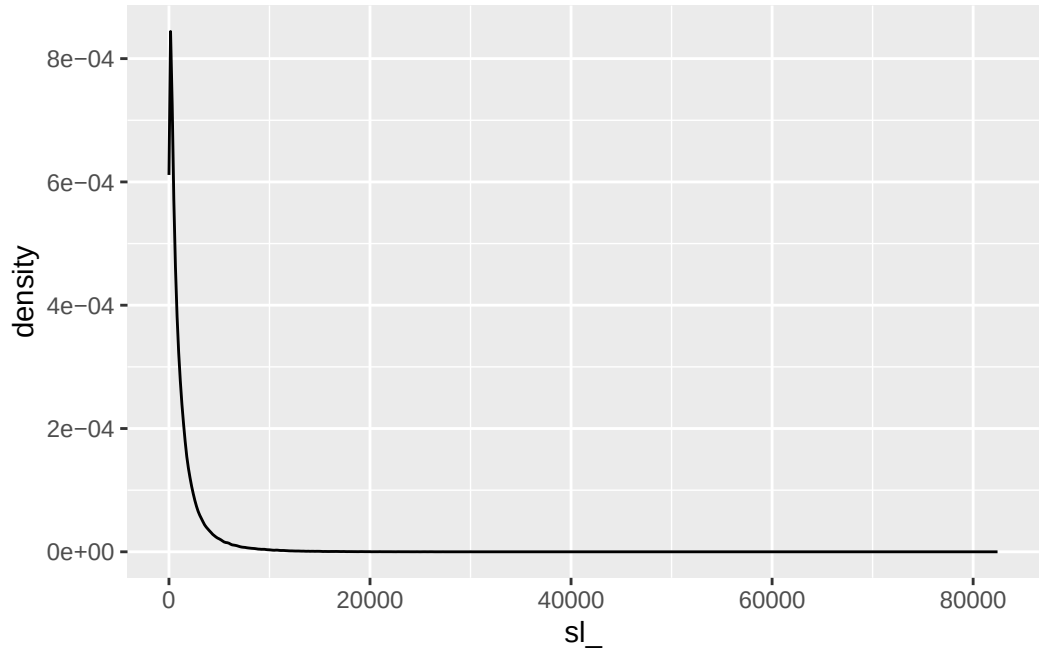
However, this fix rate leads to us only having 224397 out of the original 1223218 observations. Not terrible, but let's check another fix rate.

### 13 hour fix rate

The 75% quantile of median fix rates was 13 hours, so we'll try that next.




Let's check the step length distribution:




Again, this looks good but has some unrealistically long step lengths, so let's explore deeper.

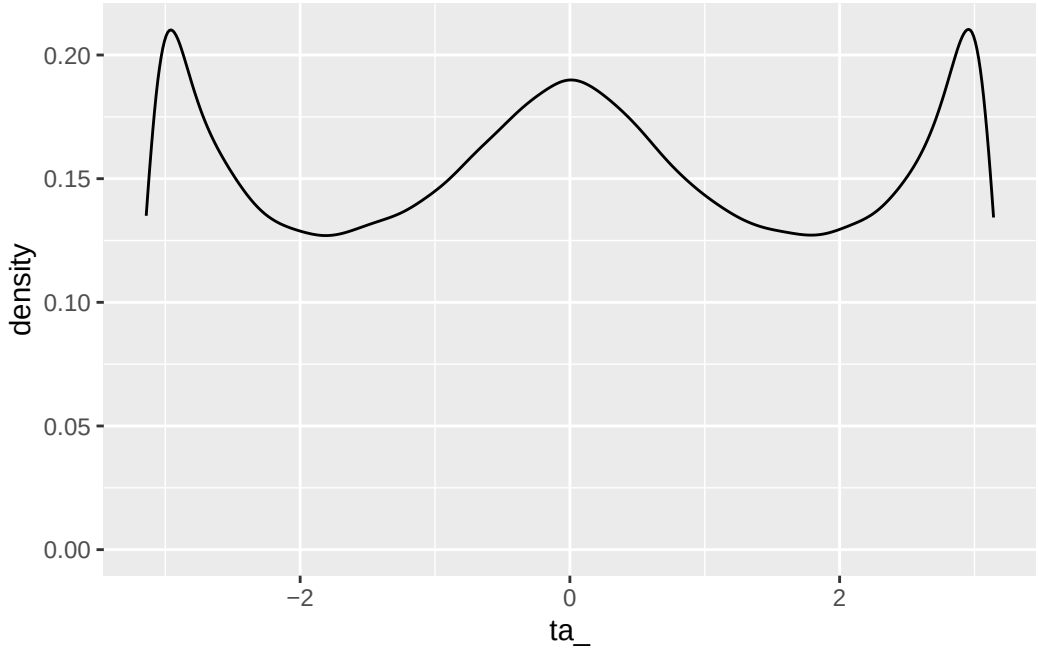
```
      0%        25%        50%        75%       100%
  0.0000   252.7824   707.2508  1684.1485 82427.1468
```

And, at the extremes:

```
     90%       95%       99%
3401.805  5088.366 10092.753
```

Again, looks like we could filter out the top 99% of steps if we need to remove the unrealistically long 66 km step taken in within 13 hours…

Let's check the turn angles:

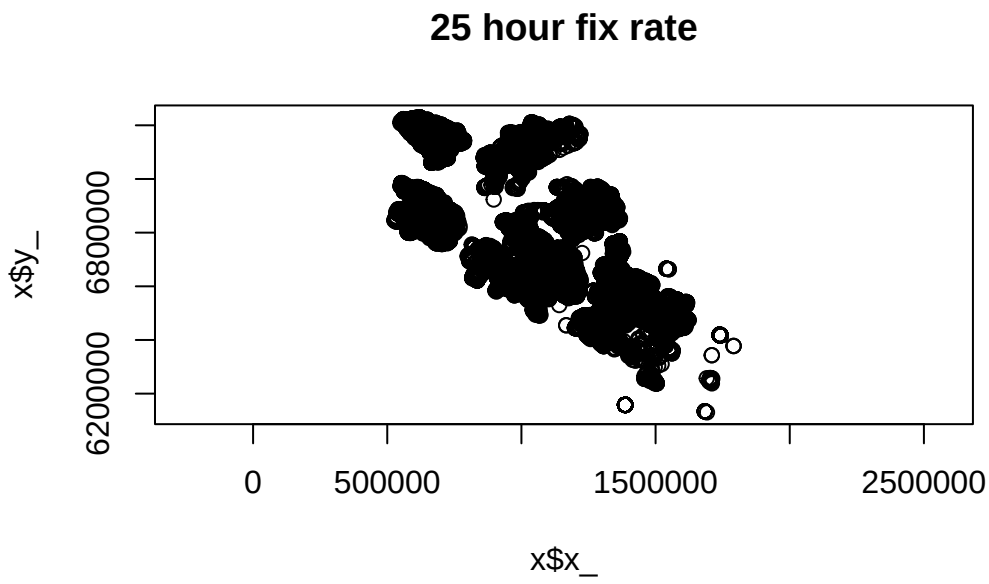



This also looks good.

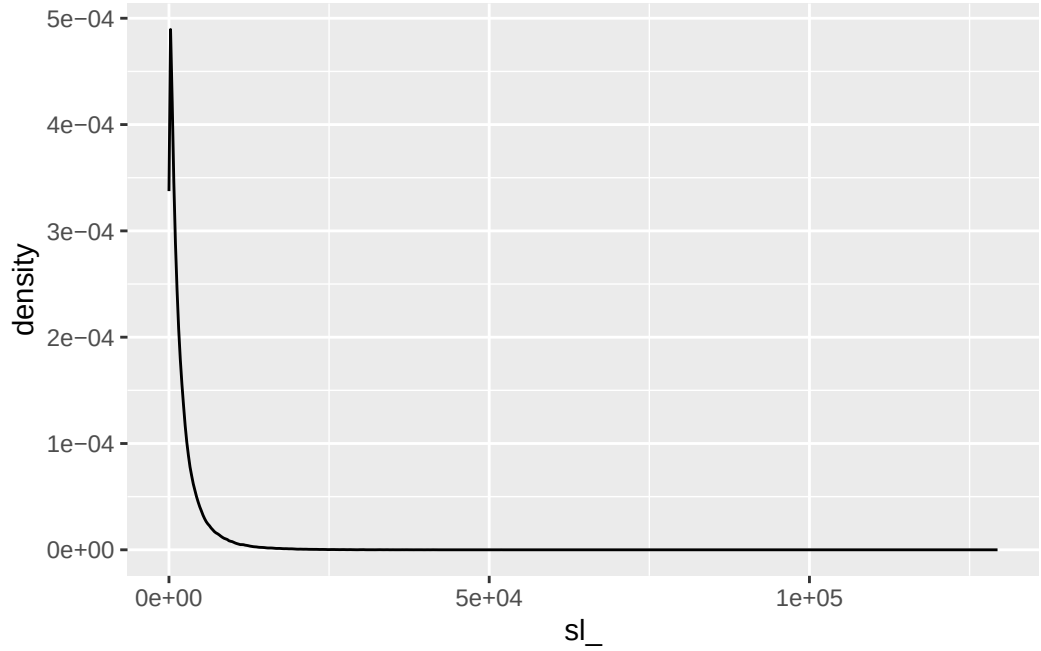
Now, this fix rate leads to us having 507417 out of the original 1223218 observations. So, this is better than the 224397 observations at the 8 hour fix rate.

### 25 hour fix rate

Let's try a 25 hour fix rate to see if we can get more data. This comes with the trade off of our inferences from movement parameters being very coarse.




Let's check the step length distribution:




Again, this looks good. Let's explore deeper.

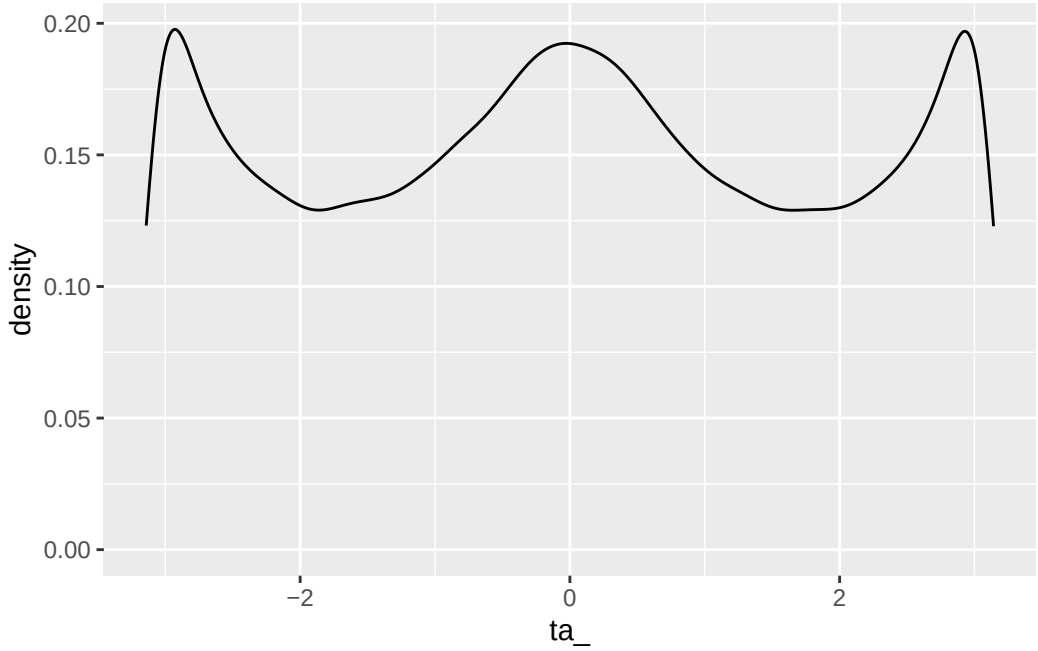
```
      0%         25%         50%         75%        100%
  0.0000    448.4574   1218.8832   2784.8482 129362.9226
```

And, at the extremes:

```
     90%       95%       99%
5424.577  7882.267 14914.276
```

Again, looks like we could filter out the top 99% of steps if we needed. Is 80 km unrealistically long in 25 hours? Probably?

Let's check the turn angles:




That's a nice looking von Mises distribution, so looks good.

Now, this fix rate leads to us having 311242 out of the original 1223218 observations. So, this is better than the 224397 observations at the 8 hour fix rate and less than the 507417 observations at the 13 hour fix rate. The 13 hour fix rate is probably our sweet spot